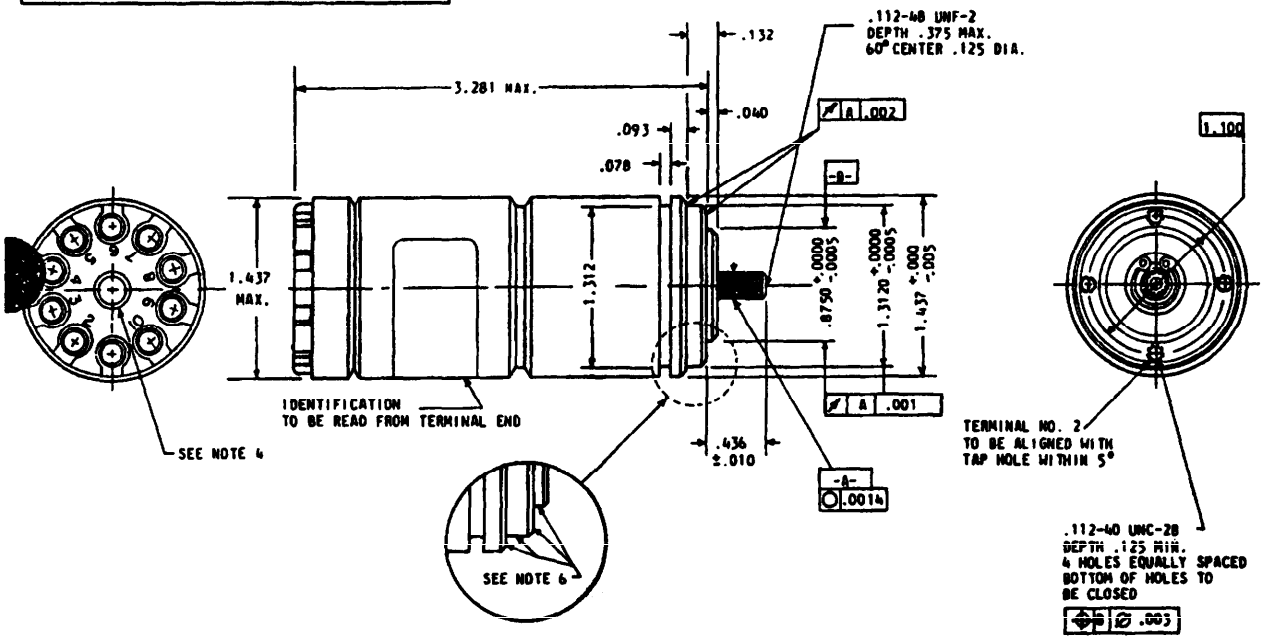
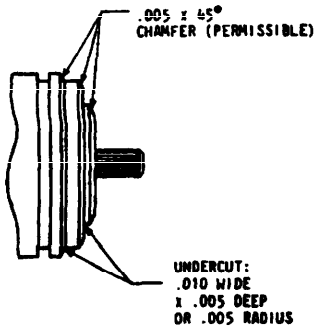
MILITARY SPECIFICATION SHEET
SERVOMOTOR – TACHOMETER GENERATOR, TYPE 15SM-RG4A

THIS SPECIFICATION SHEET IS APPROVED FOR USE BY THE AIR SYSTEMS COMMAND, DEPARTMENT OF THE NAVY, AND IS AVAILABLE FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE COMPLETE REQUIREMENTS FOR PROCUREMENT OF THE SERVOMOTOR-TACHOMETER GENERATOR DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE OF SPECIFICATION MIL-S-22820.

SPUR GEAR DATA	
TEETH	15
PITCH	96
OUTSIDE DIAMETER	.1765 $^{+.0000}_{-.0002}$
PRESSURE ANGLE	20°
TOOTH FORM	FULL DEPTH INVOLUTE
DESIGN TO AGMA 207.06 (ANSI B6.7-1967) "TOOTH PROPORTIONS FOR FINE PITCH INVOLUTE SPUR AND HELICAL GEARS." TESTED TO AGMA 390.03 "AGMA GEAR HANDBOOK, VOL. 1, GEAR CLASSIFICATION, MATERIALS AND MEASURING METHODS FOR UNASSEMBLED GEARS."	





NOTES:

1. RUNOUT SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING SHALL BE ROTATED.
2. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1 POUND RADIAL PLAY - 1/2 POUND.
3. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES ON FRACTIONS ±1/64 DECIMALS ±.005 ANGLES ±1°.
4. THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED AND INDEXED WHEN TERMINAL SCREWS ARE REMOVED.
5. THE END CAP DESIGN SHALL BE RECESSED FOR TERMINAL LUGS TO BE OF WIDTH AND DEPTH TO PREVENT TURNING OF LUGS.
6. FINISH OF HOUSING TO BE FLAT BLACK EXCEPT FOR SURFACES MARKED "X" WHICH SHALL BE BRIGHT METALLIC FINISH.
7. DIMENSIONS AND TOLERANCES SHALL COMPLY WITH ANSI-Y14.5 1973.
8. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL), A SAMPLE OF THE SERVOMOTOR-TACHOMETER GENERATOR TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE FIRST ARTICLE TESTS AS SPECIFIED IN MIL-S-22820.

FIGURE 1

FSC 6105

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TABLE 1. Performance requirements and characteristics of the servomotor section for type 15SM-RG4A servomotor-tachometer generator.

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	Hz	
Test	400	Hz	±1%
Range	400	Hz	±5%
Voltage:			
Reference Phase, Rated	115	volts	
Reference Phase, Test	115	volts	±1%
Reference Phase, Range	115	volts	±10%
Control Phase, Rated (Series)	115	volts	
Control Phase, Test (Series)	115	volts	±1%
Control Phase, Range (Series)	0 to 126.5	volts	
Control Phase, Test (Parallel)	57.5	volts	±1%
Control Phase, Range (Parallel)	0 to 63.25	volts	
Current:			
Reference Phase	.121	amperes	maximum
Control Phase (Series)	.121	amperes	maximum
Control Phase (Parallel)	.242	amperes	nominal
Power:			
Reference Phase	6.7	watts	maximum
Control Phase (Series)	6.7	watts	maximum
Control Phase (Parallel)	6.2	watts	nominal
Synchronous Speed	6000	rpm	
Impedance:			
Control Phase (Series)	525+j910	ohms	nominal
Control Phase (Parallel)	124+j229	ohms	nominal
Power Factor:			
Reference Phase	.49		nominal
Control Phase	.49		nominal
Effective Resistance:			
Control Phase (Series)	2200	ohms	maximum
Control Phase (Parallel)	550	ohms	maximum
Minimum Starting Voltage	3	volts	maximum
Coupling in Control Winding	.7	ratio	minimum
Mechanical:			
No Load Speed	4500	rpm	minimum
Stall Torque at 23°C	1.45	oz.-in.	minimum
Rotor Moment of Inertia	6.5	$gm\text{-}cm^2$	maximum
Radial Play	.0008	inches	maximum
End Play	.002-.0045	inches	min.-max.
Temperature Rise	75	degrees C	maximum
Time Constant	.032	seconds	nominal
Reversing Time	.054	seconds	nominal
Theoretical Acceleration at Stall	14,700	$radian/sec^2$	nominal
Maximum Power Output	1.3	watts	nominal

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TABLE II. Performance requirements and characteristics of the tachometer-generator section for type 15SM-RG4A servomotor-tachometer generator.

Requirements	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	Hz	
Test	400	Hz	±1%
Range	400	Hz	±5%
Voltage:			
Energizing Winding, Rated	115	volts	
Energizing Winding, Test	115	volts	±1%
Energizing Winding, Range	115	volts	±10%
Current:			
Energizing Winding	.073	amperes	maximum
Power:			
Energizing Winding	5.4	watts	maximum
Power Factor:			
Energizing Winding	.64	-----	nominal
Effective Resistance:			
Energizing Winding	2470	ohms	nominal
Impedance:			
Energizing Winding	1012+j1210	ohms	nominal
Output Winding	2200	ohms	nominal
In-Phase Speed Sensitive Transformation Ratio (at 3000 rpm) (See Note 3)	.0806	ratio	±3%
Phase Shift (3000 rpm) (See Note 3)	4.5 (lag)	degrees	nominal
In-Phase Linearity (See Note 3)	±0.8	percent	
In-Phase Axis Error (See Note 3)	.008	volts	maximum
Zero Speed Output Voltage (See Note 3)			
Total	.013	volts	maximum
Fundamental	.008	volts	maximum
Output Gradient	3.1	volts/1000 rpm	nominal

NOTES:

1. The 15SM-RG4a was formerly known as the Mk 12 Mod 0 according to BuOrd drawing 675074.

2. Phase shift changes 1/8° leading per degree C rise in ambient temperature.

3. Measured open circuit for values specified in the Table.

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NOTES: - Continued

4. The servomotor-tachometer generator identified and specified by this specification shall conform to all requirements, and shall be subjected to all tests delineated in MIL-S-22820 with the following exceptions:

a. Salt Atmosphere Resistance.
b. Audible Noise - Structureborne.
c. Voltage Sensitivity (Tachometer Section).
d. Frequency Sensitivity (Tachometer Section).
e. Temperature Sensitivity (Tachometer Section).
f. Warm-Up Time.
g. Position Error Voltage.
h. Stall Torque at High Ambient Temperature.

5. In addition to the performance requirements listed in Tables I and II, the servomotor-tachometer generator shall possess the nominal characteristics listed therein.

6. Type 15SM-RG4A servomotor-tachometer generators shall be capable of operation in ambient temperatures ranging from -55°C to 85°C.

7. The speed range for the 15SM-RG4A is 0 to 5,500 rpm. The calibration speed for determination of linearity requirements shall be 3000 rpm. Phase shift and transformation ratio tests shall be conducted at calibration speed.

8. If, during measurement, the ambient temperature is in the range of temperatures defined by 23° $\pm$ 5°C, the corrected transformation ratio shall be:

$$TR_C = \frac{TR_O}{1 - \frac{(t_O - 23)}{400}}$$

Where: TR_C = Corrected transformation ratio.
TR_O = Measured transformation ratio.
t_O = Ambient temperature at which TR_O was measured.

The corrected transformation ratio shall be within the limits specified in Table II herein. Similarly, the corrected lagging phase shift angle shall be:

$$\phi_C \text{ degrees} = \phi_O \text{ degrees} - 1/8\ (t_O - 23) \text{ degrees}$$

Where: ϕ_C degrees = Corrected lagging phase shift angle.
ϕ_O degrees = Measured lagging phase shift angle.
t_O = Ambient temperature at which ϕ was measured.

The corrected lagging phase shift shall be within the limits specified in Table II herein.

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NOTES: - Continued

9. Connection diagram:

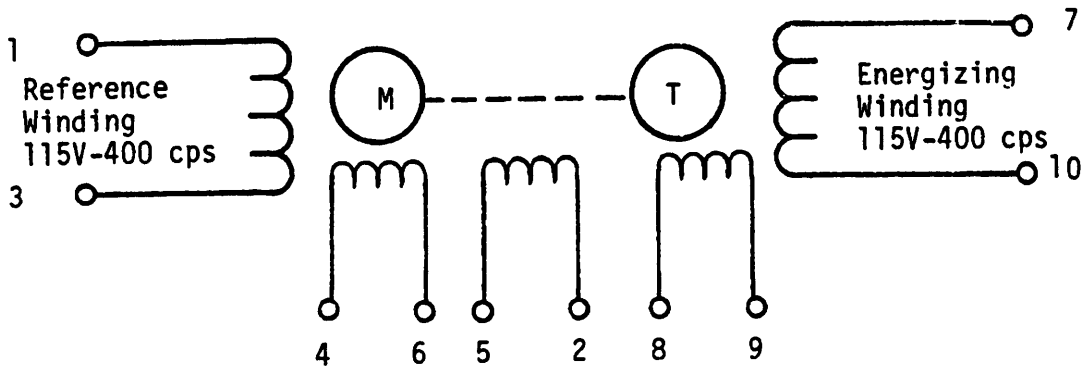


FIGURE 2. Control winding excitation 400 cps

a. For CCW rotation, the voltage at terminal 2 leads the voltage at terminal 1 by 90 electrical degrees.

b. For series operation of control winding, terminals 5 and 6 are strapped and 115 volts rated are applied across terminals 2 and 4.

c. For parallel operation of control winding, terminals 6 and 2 are strapped, terminals 4 and 5 are strapped, and 57.5 volts rated is applied across terminals 2 and 4.

d. When the shaft is rotated CCW and terminals 9 and 10 are common, the output voltage at terminal 8 is out of phase with the voltage at terminal 7.

10. Bearings shall be barrier coated in accordance with MIL-STD-1334(AS).

11. The weight of the 15SM-RG4A servomotor-tachometer generator shall not exceed 14 ounces maximum.

12. Sampling, inspection and preparation for delivery shall be in accordance with MIL-S-22820.

13. Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing Activity:
Navy AS

Project No. 6105-N120

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-22820/1B(AS) Servomotor-Tachometer Generator, Type 153M-RG4A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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